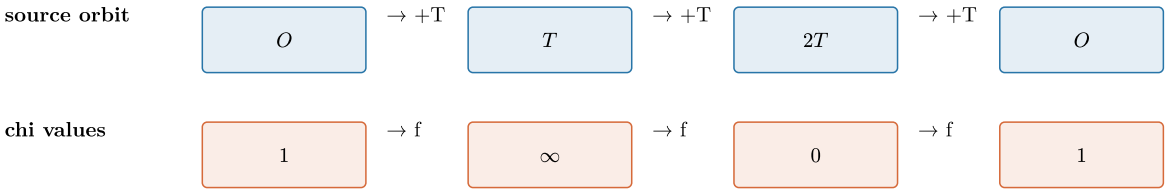
Run: cubic torsion-equivariant coordinate

2026-10-14 05:51 UTC. Protocol commit `b9f3259739`. The native Rust verifier checked the proposed `chi(P+T)=1/(1−chi(P))` identity on all rational points of four source curves and checked the order-three projective action. All assertions passed. An initial replay failed when the temporary binary was absent. After recompilation, raw and log files matched the original byte-for-byte.

p	r	Rational points checked
617	103	618
2,633	439	2,634
9,677	1,613	9,678
38,561	6,427	38,562

A cubic coordinate that distinguishes the three kernel points

For each `p` in 617, 2633, 9677, 38561: source `E/F_p:y²=x³+1`, destination `E'/F_p:y²=x³−27`, `T=(0,1)`, and the verified degree-three quotient `phi:E→E'`.



`chi=(y+1)/(y−1)` and `f(z)=1/(1−z)`; the three distinct values show that `chi` does not factor through `phi`. In contrast, `phi(O)=phi(T)=phi(2T)=O` on the destination. The finite verifier checked `chi(P+T)=f(chi(P))` on all 51,492 rational source points across the four fields and checked `f³=id` on all 51,492 projective-line points.

Source, binary, raw hashes and exact commands are in `README.md`. The coordinate does not factor through the isogeny because it takes three distinct values on the three kernel points. This is an exact coordinate check, without a decomposition oracle or complete DLP cost.